



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme



PAPER: Parametric and Nonparametric Tests (Theory)
Course Code: STAT-301

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

1. The formula for Kruskal Wallis test is based upon
(a) Means (b) Deviations (c) Ranks (d) Categories
2. Which of the following tests would you use to assess whether there is a significant difference between the mean ranks of two conditions?
(a) Kruskal Wallis (b) Spearman's Rho (c) Wilcoxon (d) Friedman
3. Nature of data for sign test is
a) Nominal or Ordinal b) Two sets of measurements can be matched
c) Direction of changing given d) All of the above
4. Non parametric tests are used on which of the following situations?
a) Sample Size is quite small b) Assumptions like normality not satisfied
c) Data is given in either nominal or ordinal scale d) All of the above
5. Which test is non sensitive?
a) None parametric b) Parametric
c) Both (a) and (b) d) None of these
6. The range of chi-square variable is
a) $0 \leq \chi^2 \leq +\infty$ b) $-\infty \leq \chi^2 \leq 0$
c) $-\infty \leq \chi^2 \leq +\infty$ d) None of these
7. To find the goodness of fit, the test used is
a) Student's t-test b) F-test
c) Chi-square test d) None of the above
8. The range of t-Statistic is
a) $0 \leq t \leq +\infty$ b) $-\infty \leq t \leq +\infty$
c) $-\infty \leq t \leq 0$ d) None of above
9. A passing student is failed by an examiner, it is an example of
a) Type I error b) Type II error c) Best decision d) All of the above
10. Homogeneity of several variances can be tested by
(a) Bartlett's test (b) Fisher's exact test (c) F test (d) T-test.



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Fifth Semester 2017
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PAPER: Parametric and Nonparametric Tests (Theory) TIME ALLOWED: 2 hrs. & 30 mins.
Course Code: STAT-301 MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q1. Write short answers on the following questions

- What is estimation? Write properties of a good estimator.
- Write the procedure of a non-parametric test which is equivalent to paired samples T-test.
- What is the Power of the test
- Write the procedure of Bartlett test
- Steps of tests of hypothesis

(4+4+4+4+4)

Q2. In a certain experiment to compare two types of sheep food A and B, the following results of increase in weights were observed:

Sheep No.	1	2	3	4	5	6	7	8
Food A	49	53	51	52	47	50	52	53
Food B	52	55	52	53	50	54	54	53

- Assuming that the two samples of sheep are independent, can we conclude that food B is better than food A?
- Examine the case when the same set of eight sheep were used in both the foods.

(8)

Q3. In an experiment to determine which of three different missile systems is preferable the propellant burning rate is measured. The data after coding, are given below. Use Kruskal Wallis test and a significance level of 0.05 to test the hypothesis that the propellant burning rates are the same for the three missile system.

(10)

Missile System		
1	2	3
24, 16.7, 22.8, 19.8, 18.9	23.2, 19.8, 18.1, 17.6, 20.2, 17.8	18.4, 19.1, 17.3, 17.3, 19.7, 18.9, 18.8, 19.3

Q4. Find the linear regression equation from the following data:

X	65, 50, 55, 65, 55, 70, 65, 70, 55, 70, 50, 55
Y	85, 74, 76, 90, 85, 87, 94, 98, 81, 91, 76, 74

Assuming normality, test the hypothesis

- $H_0: \beta = 0$ against $H_1: \beta \neq 0$;
- $H_0: \alpha = 23$ against $H_1: \alpha \neq 32$;

at the 0.01 level of significance.

(12)